

Continuous Random Variables

Recall that any discrete distribution is concentrated on a finite or countable number of isolated values. Conversely, *continuous variables can take any value of an interval*, (a, b) , $(a, +\infty)$, $(-\infty, +\infty)$, etc. Various times like service time, installation time, download time, failure time, and also physical measurements like weight, height, distance, velocity, temperature, and connection speed are examples of continuous random variables.

Computationally, to go from discrete to continuous we simply replace sums by integrals. It will help you to keep in mind that (informally) an integral is just a continuous sum.

Example 1. Let us discuss a random variable whose values are the heights of all people over 21 years of age. Between any two values, say 163.5 and 164.5 centimeters, or even 163.99 and 164.01 centimeters, there are an infinite number of heights, one of which is 164 centimeters. The probability of selecting a person at random who is exactly 164 centimeters tall and not one of the infinitely large set of heights so close to 164 centimeters that you cannot humanly measure the difference is remote, and thus we assign a probability of 0 to the event. This is not the case, however, if we talk about the probability of selecting a person who is at least 163 centimeters but not more than 165 centimeters tall. Now we are dealing with an interval rather than a point value of our random variable.

Note: For all continuous variables, the probability mass function (pmf) is always equal to zero that is

$$P\{X = x\} = P(x) = 0.$$

Definition: A random variable X is **continuous** if there is a function $f(x)$ such that for any $c \leq d$ we have

$$P(c \leq X \leq d) = \int_c^d f(x) dx. \quad (1)$$

The function $f(x)$ is called the **probability density function (pdf)**.

The pdf always satisfies the following properties:

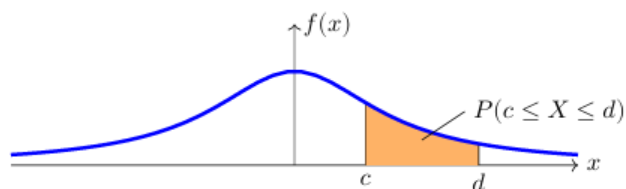
1. $f(x) \geq 0$ (f is nonnegative).
2. $\int_{-\infty}^{\infty} f(x) dx = 1$ (This is equivalent to: $P(-\infty < X < \infty) = 1$).

Note: In Property 2, we integrated over $(-\infty, \infty)$ since we did not know the range of values taken by X . Formally, this makes sense because we just define $f(x)$ to be 0 outside of the range of X . In practice, we would integrate between bounds given by the range of X .

Graphical View of Probability

If you graph the probability density function of a continuous random variable X then

$$P(c \leq X \leq d) = \text{area under the graph between } c \text{ and } d.$$



Think: What is the total area under the pdf $f(x)$?

Example 2. Suppose X has pdf $f(x) = 3$ on $[0, 1/3]$ (this means $f(x) = 0$ outside of $[0, 1/3]$). Graph the pdf and compute $P(0.1 \leq X \leq 0.2)$ and $P(0.1 \leq X \leq 1)$.

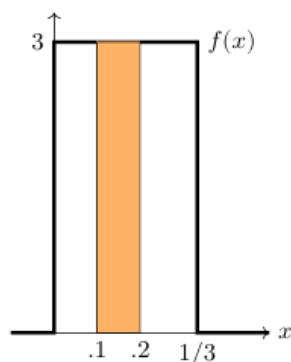
Solution: $P(0.1 \leq X \leq 0.2)$ is shown below at left. We can compute the integral:

$$P(0.1 \leq X \leq 0.2) = \int_{0.1}^{0.2} f(x) dx = \int_{0.1}^{0.2} 3 dx = 0.3.$$

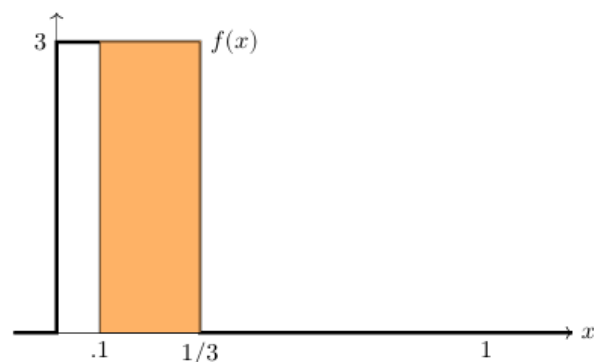
Or we can find the area geometrically:

$$\text{area of rectangle} = 3 \cdot 0.1 = 0.3.$$

$P(0.1 \leq X \leq 1)$ is shown below at right. Since there is only area under $f(x)$ up to $1/3$, we have $P(0.1 \leq X \leq 1) = 3 \cdot (1/3 - 0.1) = 0.7$.



$$P(0.1 \leq X \leq 0.2)$$



$$P(0.1 \leq X \leq 1)$$

Example 3. Let X be a random variable with range $[0,1]$ and pdf $f(x) = Cx^2$. What is the value of C ?

Solution: Since the total probability must be 1, we have

$$\int_0^1 f(x) dx = 1 \quad \Leftrightarrow \quad \int_0^1 Cx^2 dx = 1.$$

By evaluating the integral, the equation at right becomes

$$C/3 = 1 \Rightarrow \boxed{C = 3}.$$

Example 4. Let X be the random variable in the Example 3. Find $P(X \leq 1/2)$.

Solution: $P(X \leq 1/2) = \int_0^{1/2} 3x^2 dx = x^3 \Big|_0^{1/2} = \boxed{\frac{1}{8}}.$

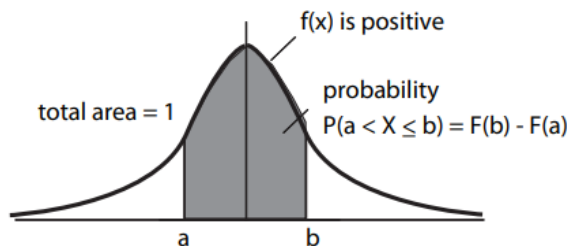
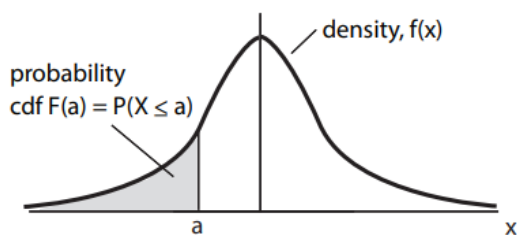
The **cumulative distribution function** $F(x)$ of a continuous random variable X with density function $f(x)$ is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt, \quad \text{for } -\infty < x < \infty.$$

Properties of cumulative distribution functions

Here is a summary of the most important properties of cumulative distribution functions (cdf)

1. (Definition) $F(x) = P(X \leq x)$
2. $0 \leq F(x) \leq 1$
3. $F(x)$ is non-decreasing, i.e. if $a \leq b$ then $F(a) \leq F(b)$.
4. $\lim_{x \rightarrow \infty} F(x) = 1$ and $\lim_{x \rightarrow -\infty} F(x) = 0$
5. $P(a \leq X \leq b) = F(b) - F(a)$
6. $F'(x) = f(x)$.



Example 3.11: Suppose that the error in the reaction temperature, in $^{\circ}\text{C}$, for a controlled laboratory experiment is a continuous random variable X having the probability density function

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Verify that $f(x)$ is a density function.
- (b) Find $P(0 < X \leq 1)$.

Solution: We use Definition 3.6.

- (a) Obviously, $f(x) \geq 0$. To verify condition 2 in Definition 3.6, we have

$$\int_{-\infty}^{\infty} f(x) \, dx = \int_{-1}^2 \frac{x^2}{3} \, dx = \frac{x^3}{9} \Big|_{-1}^2 = \frac{8}{9} + \frac{1}{9} = 1.$$

- (b) Using formula 3 in Definition 3.6, we obtain

$$P(0 < X \leq 1) = \int_0^1 \frac{x^2}{3} \, dx = \frac{x^3}{9} \Big|_0^1 = \frac{1}{9}.$$

Example 3.12: For the density function of Example 3.11, find $F(x)$, and use it to evaluate $P(0 < X \leq 1)$.

Solution: For $-1 < x < 2$,

$$F(x) = \int_{-\infty}^x f(t) \, dt = \int_{-1}^x \frac{t^2}{3} \, dt = \frac{t^3}{9} \Big|_{-1}^x = \frac{x^3 + 1}{9}.$$

Therefore,

$$F(x) = \begin{cases} 0, & x < -1, \\ \frac{x^3 + 1}{9}, & -1 \leq x < 2, \\ 1, & x \geq 2. \end{cases}$$

The cumulative distribution function $F(x)$ is expressed in Figure 3.6. Now

$$P(0 < X \leq 1) = F(1) - F(0) = \frac{2}{9} - \frac{1}{9} = \frac{1}{9},$$

which agrees with the result obtained by using the density function in Example 3.11.

Example 4.1. The lifetime, in years, of some electronic component is a continuous random variable with the density

$$f(x) = \begin{cases} \frac{k}{x^3} & \text{for } x \geq 1 \\ 0 & \text{for } x < 1. \end{cases}$$

Find k , draw a graph of the cdf $F(x)$, and compute the probability for the lifetime to exceed 5 years.

Solution. Find k from the condition $\int f(x)dx = 1$:

$$\int_{-\infty}^{+\infty} f(x)dx = \int_1^{+\infty} \frac{k}{x^3} dx = -\frac{k}{2x^2} \Big|_{x=1}^{+\infty} = \frac{k}{2} = 1.$$

Hence, $k = 2$. Integrating the density, we get the cdf,

$$F(x) = \int_{-\infty}^x f(y)dy = \int_1^x \frac{2}{y^3} dy = -\frac{1}{y^2} \Big|_{y=1}^x = 1 - \frac{1}{x^2}$$

for $x > 1$. Its graph is shown in [Figure 4.2](#).

Next, compute the probability for the lifetime to exceed 5 years,

$$\mathbf{P}\{X > 5\} = 1 - F(5) = 1 - \left(1 - \frac{1}{5^2}\right) = 0.04.$$

We can also obtain this probability by integrating the density,

$$\mathbf{P}\{X > 5\} = \int_5^{+\infty} f(x)dx = \int_5^{+\infty} \frac{2}{x^3} dx = -\frac{1}{x^2} \Big|_{x=5}^{+\infty} = \frac{1}{25} = 0.04.$$

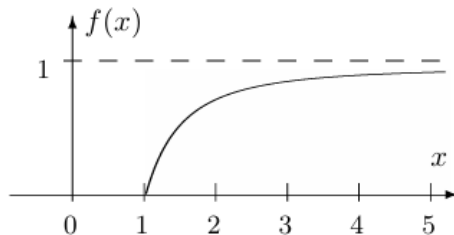


FIGURE 4.2: Cdf for Example 4.1.

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Distribution	Discrete	Continuous
Definition	$P(x) = P\{X = x\}$ (pmf)	$f(x) = F'(x)$ (pdf)
Computing probabilities	$P\{X \in A\} = \sum_{x \in A} P(x)$	$P\{X \in A\} = \int_A f(x)dx$
Cumulative distribution function	$F(x) = P\{X \leq x\} = \sum_{y \leq x} P(y)$	$F(x) = P\{X \leq x\} = \int_{-\infty}^x f(y)dy$
Total probability	$\sum_x P(x) = 1$	$\int_{-\infty}^{\infty} f(x)dx = 1$

TABLE 4.1: *Pmf* $P(x)$ versus *pdf* $f(x)$.